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Implementing Secure Genome Language Models for National Antimicrobial Resistance Surveillance

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Target Audience Analysis:

The target audience for this paper is PHAC's Executive Leadership and Technical Advisory Committee. As a highly risk-averse group, they rely on traditional alignment tools like BLAST to prevent misreporting and are hesitant to adopt Genome Language Models (GLMs) due to 'black-box' explainability concerns and genomic privacy risks. To effectively persuade them, this paper validates their skepticism as a necessary defense of public health standards. However, it will ultimately argue that the high false-negative rates of their current methods pose an even greater, hidden threat to national biosecurity, making the transition to a secure GLM framework essential.

Implementing Secure Genome Language Models for National Antimicrobial Resistance Surveillance

1. Introduction: The Global AMR Crisis and the Case for Genome Language Models

The rapid spread of Antimicrobial Resistance (AMR) represents one of the most critical threats facing modern clinical medicine and global health security. According to the 2016 O'Neill report, if current trajectories remain unchecked, the global death toll attributed to antibiotic-resistant pathogens will reach 10 million annually by 2050 [1], [2]. In addition to the human casualties, antimicrobial resistance is projected to cost the world approximately 100 trillion USD, fundamentally destabilizing global markets and existing healthcare systems [1], [2].

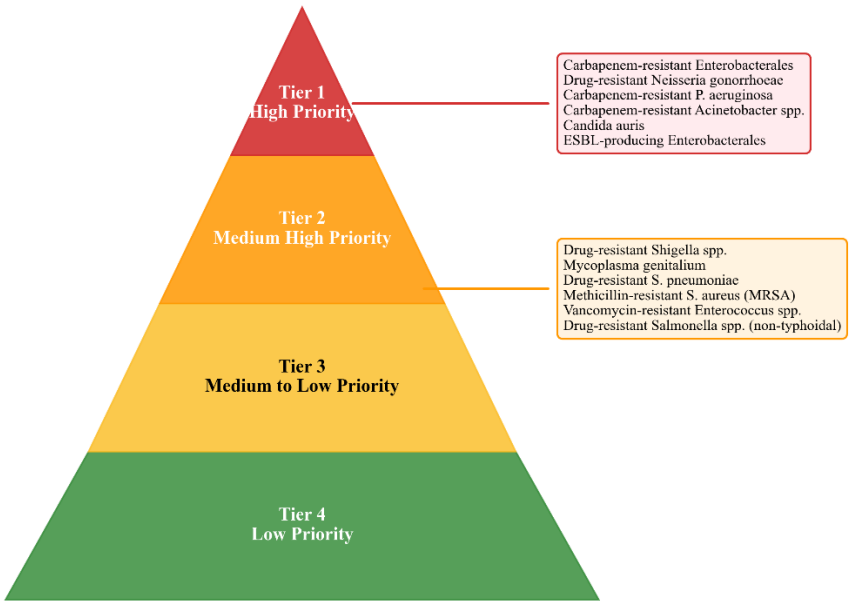


Figure 1: Canada's 2025 Priority AMR Pathogens grouped by risk tier. The most critical threats (Tier 1 and 2) require highly sensitive detection methodologies. Adapted from the pathogen multi-criteria decision analysis by Abdesselam et al. [3].

Recent real-world data demonstrates that these alarming forecasts are already materializing. In 2019 alone, bacterial AMR directly caused approximately 1.3 million deaths globally, while contributing to over 5 million deaths in total [4]. Despite the implementation of ongoing public health interventions, national surveillance data confirms that Canada's antimicrobial resistance crisis has steadily deteriorated since 2015 [3], [4]. To effectively protect public health and fulfill the 2023 Pan-Canadian Action Plan on AMR, the Public Health Agency of Canada (PHAC) needs proactive national surveillance; however, current platforms remain constrained by foundational analytical limitations [3].

Recent PHAC assessments point to structural deficits within AMRNet, noting that the platform “lacks routine genotypic testing, limiting insight into resistance mechanisms” [3]. This gap is especially alarming for high-priority carbapenemase-producing organisms (CPOs). Because "no routine molecular data is collected" for CPOs across all settings, these diagnostic blind spots directly "hinder the ability to fully characterize resistance mechanisms, assess treatment implications, and capture the breadth of transmission" [3]. Depending on these constrained pipelines leaves Canada vulnerable to undetected, novel resistance threats, highlighting the urgent need to adopt sequence-aware computational tools like Genome Language Models (GLMs) [2].

While traditional alignment algorithms remain the gold standard for identifying well-documented pathogens, they are structurally constrained by their absolute dependence on direct sequence homology [2]. Consequently, when processing high-volume, complex metagenomic data—from wastewater and agricultural runoff to polymicrobial clinical samples—these traditional pipelines routinely miss novel or highly divergent antimicrobial resistance genes (ARGs) that lack high identity with established databases [2]. This systemic failure to detect emerging threats leaves a critical and unmonitored gap in our national health tracking frameworks.

To secure national health infrastructure, public health bioinformatics must pivot from legacy sequence alignment tools toward advanced, machine learning-driven detection platforms. PHAC must strategically integrate GLMs into national AMR tracking through a highly controlled deployment model: utilizing an air-gapped, local computing environment and a two-tier verification system that pairs strict automated probability cutoffs with specialized human review. This proposed model systematically eliminates the sequence homology bottleneck in current surveillance, offering advanced detection capabilities without compromising agency standards for privacy, auditability, or clinical rigor.

2. Evaluating Technical Skepticism at PHAC

Integrating artificial intelligence with PHAC's existing pathogen tracking infrastructure requires addressing valid methodological skepticism. Adopting this hybrid approach demands justifiable caution, given well-documented concerns regarding model interpretability, data governance, and clinical reliability.

2.1 Model Opacity and Interpretability

The primary critique of deep neural networks is their opacity. Conventional pipelines rely on deterministic sequence homology alignments, yielding concrete, auditable outputs [5], [6]. In contrast, deep learning frameworks process data through highly abstract feature spaces [2], [6]. To public health officials, these models can appear as uninterpretable "black boxes" lacking clear biological traceability, complicating accountability.

2.2 Clinical Reliability and Operational Risk

Early deep learning models suffered from high false-positive rates in certain complex categories (e.g., 0.27 precision for multidrug resistance), incorrectly annotating benign sequences as multi-drug threats [2]. If deployed without oversight, these inaccuracies risk severe operational consequences, including misdirected epidemiological deployments, unnecessary hospital quarantines, and unjustified public health warnings.

2.3 Data Sovereignty and Metagenomic Privacy Risks

Furthermore, the privacy and security risks associated with analyzing unflagged metagenomic sequencing pools represent a substantial legal and ethical hurdle for national health networks. Metagenomic samples collected from wastewater treatment plants, livestock facilities, or clinical swabs inevitably contain residual human host DNA alongside microbial sequences [4]. Transferring these raw or partially filtered sequencing reads to external cloud platforms or third-party AI servers introduces severe risks of data interception, explicitly threatening data sovereignty and privacy mandates [7], [8]. To maintain public trust, modern analytical frameworks must strictly prioritize data integrity and patient confidentiality.

3. Limitations of Current Alignment Models

While PHAC's caution regarding AI is understandable, relying on legacy alignment tools poses an even greater, unquantified risk to national biosecurity. Traditional surveillance methods possess structural failures, whereas GLMs provide a necessary, computationally superior alternative.

3.1 The False-Negative Crisis in Traditional Alignment

Canada's public health infrastructure heavily relies on traditional sequence alignment algorithms for genomic surveillance. While these tools remain the gold standard for sequence analysis and work effectively for identifying well-documented resistance genes, their strict identity cutoffs introduce a severe operational blind spot, driving high false-negative rates for novel threats [2], [9]. Consequently, current infrastructure relies on a constrained definition of resistance biased toward historically known genes. If an emerging threat possesses under 40% sequence homology to indexed databases, standard best-hit tools consistently fail to detect it [2]. Therefore, relying on these rigid alignment pipelines actively undermines PHAC’s mandate to proactively identify novel resistance threats.

3.2 Computational and Biological Bottlenecks of Legacy Algorithms

The core bottleneck lies in both the computational and biological limitations of these legacy algorithms. Because exact multiple sequence alignment (MSA) algorithms suffer from exponential computational complexity, executing optimal alignments has become prohibitively

slow and resource-intensive for today's massive, high-throughput genomic databases [10]. Furthermore, because alignment-based pipelines inherently struggle to process large structural variations and horizontal gene transfer (HGT) events, they remain effectively blind to highly divergent resistance mechanisms and virus-host interactions that share minimal sequence homology [9]. As a direct consequence, public health agencies risk overlooking critical environmental 'hot spots' for genetic exchange, allowing undetected resistance mechanisms to circulate freely across environmental reservoirs and ultimately precipitate crises in clinical settings.

3.3 Detecting Novel Threats Beyond Strict Sequence Homology

Deep learning models like DeepARG bypass the strict cutoffs and best-hit limitations of traditional pipelines by converting initial alignment scores into a comprehensive feature matrix [2]. Rather than matching exact sequence strings, these tools utilize dissimilarity matrices that measure distances relative to all known ARG categories simultaneously [2].

An independent assessment by Berglund et al., detailing 76 novel metallo-beta-lactamase determinants, underscores the critical need to adopt AI-driven surveillance architectures [2]. Experimental validation confirmed that the vast majority of these novel candidates successfully conferred phenotypic resistance to carbapenems, severely compromising a critical class of last-resort antibiotics [2]. Because these sequences shared under 40% identity with cataloged database entries, standard alignment tools failed to detect them [2]. By contrast, utilizing high-dimensional sequence patterns allowed the DeepARG-LS architecture to successfully identify 85% of these divergent sequences as severe beta-lactamase threats [2]. Relying solely on legacy alignment methods allows divergent, uncatalogued resistance genes to spread unchecked across environmental networks.

3.4 The Next Frontier: Alignment-Free Genome Language Models

GLMs represent a new paradigm eliminating explicit sequence alignment. By extracting context-aware representations directly from raw genetic data, GLMs capture the underlying structural, functional, and evolutionary characteristics of resistance mechanisms [6], [11]. Ultimately, this shift toward high-dimensional genomic pattern analysis fundamentally eliminates the agency's reliance on strict sequence identity matching.

Recent architectural innovations, notably the HyenaDNA framework, radically expand these analytical capabilities by evaluating massive context windows of up to one million tokens while maintaining precise single-nucleotide resolution [12]. This unprecedented scale is highly advantageous because many complex resistance mechanisms and HGT events rely on ultra-long-range structural dependencies that span hundreds of thousands of base pairs [11], [12]. By evaluating massive bacterial genomic neighborhoods simultaneously, these models can identify distant regulatory elements and hidden operon structures that traditional, fragmented pipelines routinely lose [11], [12]. Ultimately, this guarantees that PHAC’s surveillance infrastructure can detect both large-scale evolutionary shifts and the subtle single-nucleotide mutations that trigger broad-scale clinical resistance.

3.5 Demystifying the Black Box via Explainable AI

For public health leadership, the perceived 'black-box' architecture of deep neural networks represents a principal regulatory and clinical barrier to GLM integration. However, recent breakthroughs in Explainable AI (XAI) successfully demonstrate that Transformer-based genome language models can be rendered highly transparent and clinically interpretable [6], [13]. Through the adaptation of Layer-wise Relevance Propagation for attention mechanisms (AttnLRP), algorithmic outputs from models like DNABERT-2 can now be directly traced to specific biological motifs at the single-nucleotide level [13]. By verifying that AI predictions correspond to established biological annotations, this framework provides the exact mechanistic visibility and auditability required by clinical advisory committees.

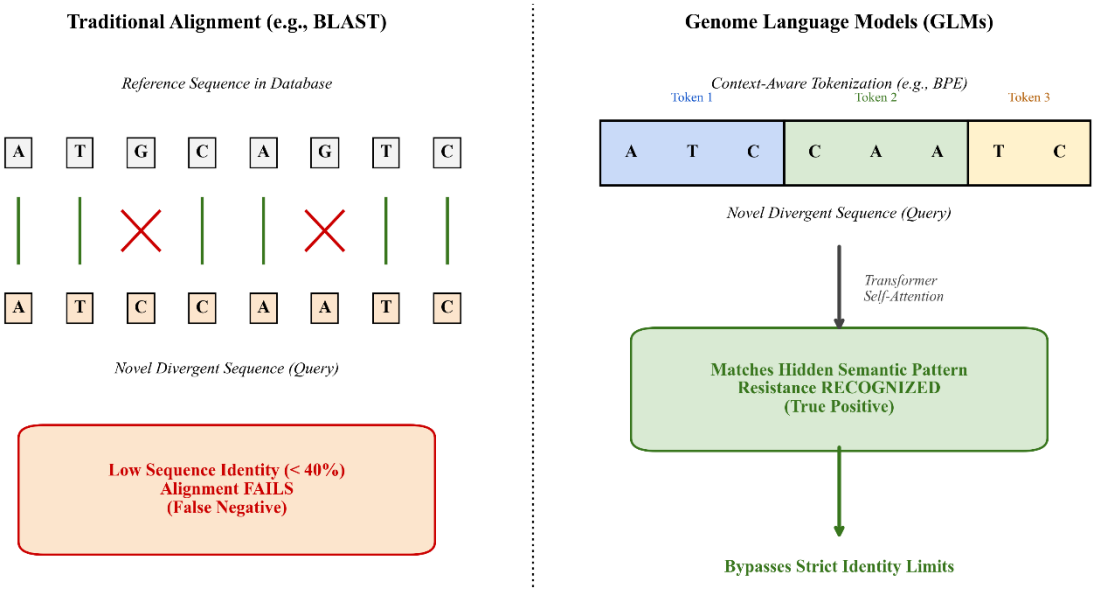


Figure 2: Comparison of traditional homology-based alignment and GLM tokenization. Traditional alignment methods (left) rely on strict base-by-base matching and fail to detect divergent variants. In contrast, GLMs (right) utilize context-aware subword tokenization to capture underlying structural and functional semantics, successfully identifying novel resistance mechanisms. Tokenization models adapted from the architectural review by Shu et al. [11].

4. Proposed Framework Part I: The Two-Tier Verification System

To balance high-sensitivity detection with operational precision, PHAC must implement a Two-Tier Verification System, inspired by the multilayered governance frameworks for genetic AI proposed by Sokhansanj and Rosen [5]. This hybrid architecture harnesses the advanced pattern-recognition capabilities of GLMs to detect highly divergent resistance mechanisms. To rigorously control false-positive rates, the system enforces the probability cutoffs and manual expert review protocols established by Arango-Argoty et al [2].

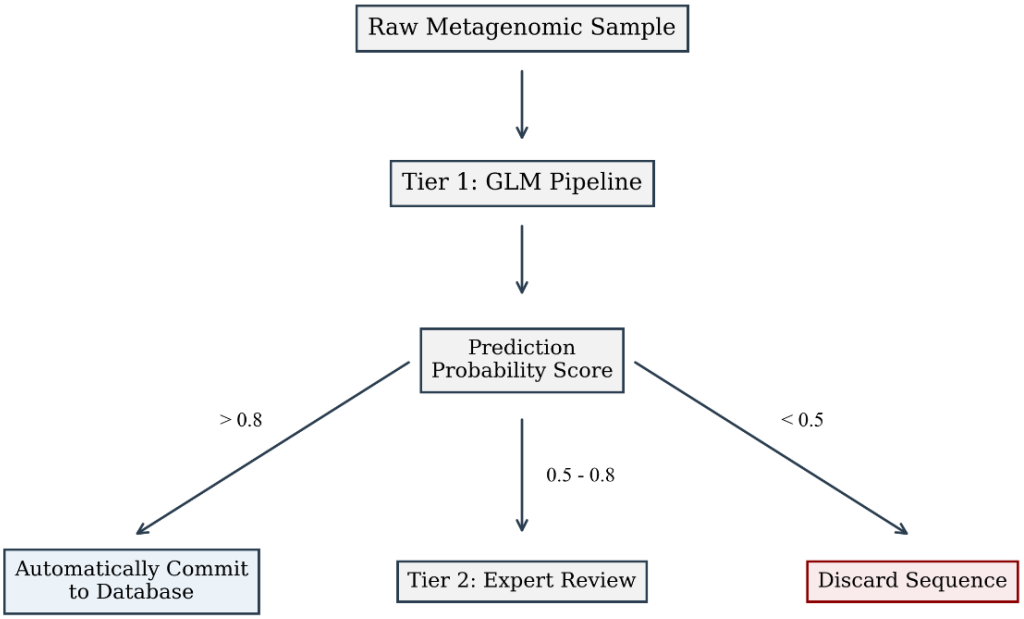


Figure 3: Proposed Two-Tier Verification Workflow for GLM Predictions. The automated probability cutoffs and manual inspection routing are adapted from the sequence classification thresholds established by Arango-Argoty et al. [2].

4.1 Tier 1: Automated Deep Learning Screening

Tier 1 leverages optimized GLMs—specifically DNABERT-2 for short-read sequences and HyenaDNA for ultra-long contexts—executing rapid genomic triage. HyenaDNA's million-token capacity captures ultra-long-range structural dependencies and mutational profiles missed by legacy tools [12]. To mitigate over-reporting concerns, this pipeline enforces a strict 0.8 prediction probability cutoff, ensuring only verified, high-confidence predictions enter the national registry [2].

4.2 Tier 2: Specialized Human-in-the-Loop Validation Protocol

Predictions falling between the 0.5 and 0.8 confidence thresholds are routed to Tier 2 for manual inspection. Expert analysts use AttnLRP explainability frameworks to visually trace the nucleotide positions driving the GLM's classification. This human oversight prevents misclassification errors while capturing highly divergent resistance genes that legacy tools overlook.

4.3 Statistical Performance and Operational Rigor

Across comprehensive benchmark testing, modern GLMs demonstrate the operational efficiency and analytical rigor required for national public health surveillance. Specifically, DNABERT-2 achieves state-of-the-art predictive performance while utilizing 21 times fewer parameters and approximately 92 times less computational pre-training time than previous foundational models [14]. Similarly, HyenaDNA establishes new accuracy standards across multiple genomic benchmarks, outperforming traditional convolutional neural networks by up to 20 accuracy points in regulatory element identification [12]. Ultimately, by extracting complex sequence features across vast context windows instead of relying on rigid sequence homology, this proposed Two-Tier system delivers an optimal balance of rapid automated processing, high-sensitivity detection, and critical expert oversight.

5. Proposed Framework Part II: Air-Gapped Local Architecture and Privacy Controls

Deploying advanced AI models within national public health infrastructure demands rigorous operational security and data governance. To satisfy federal privacy mandates protecting sensitive genomic information, PHAC must implement the federated, *in situ* analysis framework conceptualized by Casaletto et al. [8], reinforced by the hybrid cryptographic access controls proposed by Oghenekevwe and Adawaren [7]. This dual architecture ensures that AI processing remains strictly confined to isolated computing environments while actively preventing the external transmission of raw data. Ultimately, by running the AI algorithms directly on local hospital servers and enforcing fine-grained decryption policies rather than transmitting sensitive patient DNA to the cloud, PHAC can harness the predictive insights of GLMs without compromising patient confidentiality or national data sovereignty.

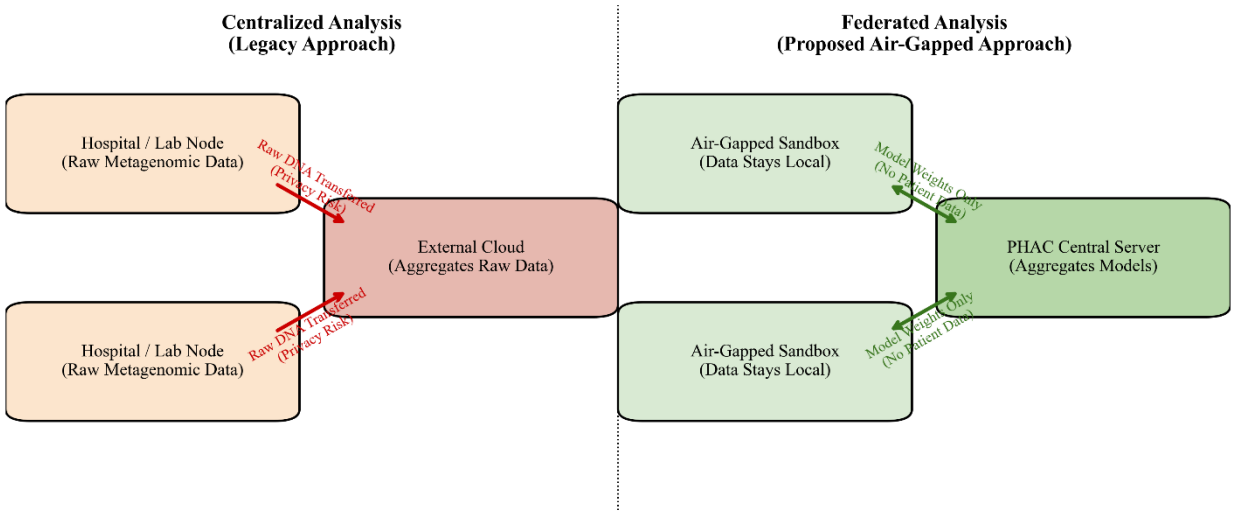


Figure 4: Comparison of Centralized versus Federated data analysis architectures. In the proposed Federated model, raw metagenomic data remains secured in local air-gapped sandboxes, and only non-identifiable model weights are transmitted. Adapted from the federated computing architectures described by Casaletto et al. [8].

5.1 Infrastructure Architecture and Localized Computation

A perceived barrier to GLMs is their dependence on massive GPU clusters and cloud connectivity. However, modern GLMs like DNABERT-2 and HyenaDNA are highly optimized for efficient inference, requiring significantly less computational power than earlier models [12], [14]. This unprecedented computational efficiency empowers PHAC to securely deploy the entire analytical pipeline locally within an 'air-gapped' architecture. Furthermore, because agencies can leverage open-source foundational weights, localized model training (fine-tuning) can be executed using parameter-efficient methods like Low-Rank Adaptation (LoRA) or soft-prompting [11], [12], [14]. These techniques reduce the trainable parameters to a fraction of their original size, allowing PHAC to continuously update the models on standard, local agency hardware without requiring massive cloud-computing budgets. Consequently, this localized framework completely circumvents external data transmission risks and the inherent vulnerabilities of third-party cloud vendors

5.2 Privacy-Preserving Federated Learning

To safely contribute to global AMR intelligence networks without violating privacy mandates, PHAC should implement federated learning. This framework exchanges only learned model parameters, keeping raw metagenomic sequence reads permanently confined to local servers and preventing the centralization of sensitive human DNA [6], [8]. Furthermore, to safely share these single-nucleotide insights, PHAC can pair GLM outputs with Secure Multi-Party Computation (SMC) and Attribute-Based Encryption (ABE) [7]. SMC enables joint epidemiological computations over private inputs, while ABE enforces fine-grained access control, ensuring PHAC satisfies strict global privacy regulations like HIPAA and GDPR without disclosing raw data.

5.3 A Low-Risk Deployment Strategy for Real-Time Surveillance

To successfully modernize public health frameworks, agencies must practically assess their operational capabilities and resource efficiencies. By operating pre-trained GLMs locally, the agency bypasses the severe computational bottlenecks of traditional alignment, creating a highly efficient, secure environment for real-time pathogen tracking [15]. Importantly, this localized infrastructure operates synergistically with PHAC's trusted reference databases, augmenting tools like CARD and ResFinder instead of discarding them [2].

Through a hybrid approach, PHAC will continue utilizing its legacy alignment pipelines to ensure high-confidence confirmation of well-documented clinical threats. The GLM framework will be leveraged for its zero-shot predictive power, allowing the agency to accurately identify emerging, highly divergent resistance genes without needing prior training examples or homologous references [12]. By operating these systems in tandem, the agency achieves a comprehensive biosecurity posture—capable of detecting both documented pathogens and evolutionary mutations—without disrupting trusted clinical routines.

6. Conclusion: Strategic Path Forward

To combat the escalating crisis of AMR, the PHAC must modernize its biosecurity infrastructure by moving beyond traditional alignment methods that leave the agency dangerously blind to divergent, novel pathogens. Adopting an advanced GLM framework permanently closes this fatal blind spot, and this transition can be achieved safely through a conditional adoption framework. By implementing a Two-Tier Verification System that pairs rapid automated GLM screening with rigorous human oversight, the agency can guarantee analytical precision while effectively controlling false positives. Furthermore, deploying these models within an air-gapped environment ensures uncompromising data privacy, directly mitigating the committee's security concerns. Because this GLM infrastructure is designed to integrate seamlessly alongside established diagnostic tools like BLAST and CARD, it elevates national surveillance capabilities without disrupting existing clinical workflows. Ultimately, by embracing this highly controlled and transparent AI strategy, PHAC will overcome the fatal limitations of legacy systems and proactively defend public health against tomorrow's undetected superbug threats.

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